

AI Teacher: Human-Like AI Educator

Complete Technical Project Documentation & System Architecture Specification

Challenge: Build a Human-Like AI Educator That Teaches Through Video	Event: AI Innovation Hackathon 2026 (Round 2)
Team: AI Cognitive Creators	Tech Stack: FastAPI, React 18, Vite, Groq Llama 3.3, Edge-TTS, OpenCV

1. Problem Statement

Traditional digital learning platforms suffer from two fundamental bottlenecks that hinder effective student learning:

- Static Pre-Recorded Video Lectures:** One-size-fits-all content where pacing cannot adjust to the learner, difficult concepts cannot be explained with tailored real-world analogies, and passive watching leads to high drop-off rates and the 'illusion of competence.'
- Text-Based AI Chatbots:** Standard conversational interfaces lack visual presence, structured pedagogical progression, vocal emotion, and dynamic whiteboard diagrams. Text-heavy chat walls cause cognitive fatigue and fail visual and auditory learners.

The Challenge: To design and develop an AI-powered Virtual Human Educator capable of ingesting educational materials (textbooks, PDFs, notes, slides) or accepting a conceptual topic directly, transforming materials into a structured pedagogical curriculum, and teaching students through synchronized talking video, expressive neural voice, and real-time whiteboard diagrams while proactively diagnosing misconceptions and answering doubts in real-time.

2. Solution Overview

The **AI Teacher Platform** is an end-to-end full-stack educational system built using a Pedagogical State Machine Graph, Retrieval-Augmented Generation (RAG), Real-Time Lip-Synced Video Synthesis, and a modern Reactive Web UI:

- **Dual Ingestion Mode:** Supports both General Knowledge Topic Mode (dynamic curriculum generation) and Document-Grounded Mode (RAG vector indexing of uploaded textbooks/notes).
- **Multimodal Video Classroom:** Synchronizes audio-driven talking educator avatars with dynamic whiteboard visual schemas and educator script captions.
- **Closed-Loop Adaptive Pedagogy:** Breaks topics into digestible micro-modules, conducts interactive checkpoints, remediates misconceptions on the fly, provides active recall flashcards, and delivers comprehensive diagnostic assessments.

3. Key Features

Category	Feature	Capability Summary
Ingestion	Dual Ingestion Modes	General Knowledge topic mode or Document-Grounded mode (PDF, DOCX, PPTX).
Ingestion	TOC-Aware Chapter Locator	Parses Table of Contents to target exact chapter titles and PDF page offsets.
Pedagogy	Adaptive Time Modes	■ 5m Blitz, ■■ 20m Standard, ■ 60m Masterclass, and ■ 7-Day Curriculum
Pedagogy	Teacher Personalities	Dr. Sarah (Intuitive), Prof. Aryan (Analytical), Coach Maya (Exam-Prep).
Pedagogy	Remediation Loops	Branches into alternate simplified analogies when student misconceptions occur.
Interactivity	In-Lesson Doubt Solver	Ask doubts via text or microphone; teacher replies in-character with voice audio.
Interactivity	Multilingual Switcher	1-click in-lesson toggle between English, Hindi, and Hinglish.
Retention	3D Flashcards & Notes	Active recall card deck with analogies plus 1-click Markdown notes export.
Roadmaps	Learning Paths	Progressive multi-stage roadmaps for broad subjects with 1-click lesson launch.

4. System Architecture

The system is organized into a decoupled, high-performance architecture:

- **Frontend Layer:** Single Page Application built with React 18, Vite, and Tailwind CSS. Employs HTML5 Video Player with HTTP 206 Partial Content range requests for smooth video playback, and Web Speech API for low-latency microphone audio capture.
- **FastAPI Backend:** Asynchronous Python 3.11 ASGI router providing REST endpoints for sessions, curriculum generation, RAG document processing, media streaming, and user authentication.
- **Orchestration Engine:** Deterministic state machine governing pedagogical transitions from planning to teaching, micro-checkpoints, remediation, and quizzes.
- **Synthesis Engines:** Edge-TTS neural speech generation, Matplotlib/Graphviz whiteboard rendering, and OpenCV facial motion synthesis.
- **Data & Persistence:** SQLite database via SQLAlchemy ORM for profiles and sessions, ChromaDB for vector storage, and an on-disk cache for media assets.

5. AI/ML Models Used

Capability	Model / Framework	Provider	Role & Description
Curriculum Planning	llama-3.3-70b-versatile	Groq Cloud	Plans lessons, spoken scripts, analogies, and checkpoint rubrics.
Fast Doubt & Eval	llama-3.1-8b-instant	Groq Cloud	Low-latency inference for doubt resolution and quiz evaluation.
Vector Embeddings	all-MiniLM-L6-v2	HuggingFace	384-dimensional dense semantic vectors for ChromaDB document search.
Speech-to-Text	whisper-large-v3	Groq Cloud	Robust speech transcription for microphone input in English & Hindi.
Neural Voice (TTS)	Edge-TTS Neural Models	Microsoft	Natural human-like narration (JennyNeural, GuyNeural, SwaraNeural).
Lip-Sync Synthesis	OpenCV + FFmpeg	Local Engine	Audio volume-envelope affine mouth warping in under 2 seconds.

6. RAG (Retrieval-Augmented Generation) Implementation

The RAG pipeline enables curriculum generation strictly grounded in course materials:

- Multi-Format Ingestion:** Supports PDF, DOCX, PPTX, TXT via PyMuPDF and python-docx.
- TOC Extraction:** Inspects document front matter to map chapter headings to physical page numbers, preventing misidentifications.
- Semantic Chunking:** Recursive character splitting with 1,000-character chunk sizes and 200-character overlaps.
- Vector Storage:** Encoded via Sentence-Transformers into ChromaDB with cosine similarity indices.
- Anti-Hallucination Grounding:** Retrieved chunks (top-k = 5) are injected into the LLM system prompt with strict rules requiring source page citations and forbidding fabricated information.

7. Prompt & Agent Architecture

The Pedagogical Engine is governed by a **Deterministic Finite State Machine (FSM)**:

INIT → PLANNING → TEACHING_CONCEPT → AWAITING_CHECKPOINT → [REMIEDIATION] →
NEXT_CONCEPT → FINAL_ASSESSMENT → COMPLETED.

Pydantic Schema Enforcement: Every model invocation validates output strictly against Pydantic schemas (e.g. LessonPlan, DoubtResponse, FlashcardDeck). A self-repair parser intercepts malformed JSON and retries with progressive fallbacks to guarantee reliability.

8. Personalization Approach

Personalization adapts across 4 distinct dimensions:

- **Cognitive Level:** Beginner (intuitive analogies), Intermediate (balanced theory/math), and Advanced (system-level trade-offs and edge cases).
- **Time Budgeting:** 5m Blitz, 20m Standard, 60m Masterclass, and 7-Day Curriculum.
- **Teacher Personas:** Dr. Sarah (empathetic/supportive), Prof. Aryan (analytical/technical), and Coach Maya (high-energy/exam prep).
- **Misconception Remediation:** Dynamically diagnoses student misconceptions and presents an alternative mental model to clear confusion.

9. Assessment Methodology

Combines formative and summative evaluations:

- **Formative Checkpoints:** Open-ended conceptual questions at the end of each module evaluated against an expected rubric, scoring from 0.0 to 1.0.
- **Summative Assessment:** 5 multi-tier questions mapped to Bloom's Taxonomy (Recall, Comprehension, Application, Analysis, Synthesis).
- **Diagnostic Learning Report:** Generates mastery rates, identified strengths, weaknesses, and personalized next-step revision guidance.

10. Multilingual Implementation

Full support for **English**, **Hindi** (Devanagari), and **Hinglish** (conversational Hindi-English blend used across Indian universities). The **Dynamic In-Lesson Switcher** allows learners to switch languages mid-lesson; the backend dynamically re-narrates the educator script, regenerates voice audio, and re-synthesizes the lip-sync video without restarting the session.

11. Voice Implementation

- **Text-to-Speech (TTS):** High-definition neural speech synthesis using Microsoft Edge-TTS with distinct voices assigned per persona (Dr. Sarah: JennyNeural, Prof. Aryan: GuyNeural / MadhurNeural, Coach Maya: SwaraNeural).
- **Speech-to-Text (STT):** Client-side Web Speech API provides zero-latency microphone transcription for asking doubts, backed by server-side Groq Whisper Large v3 for uploaded audio files.

12. Avatar & Video Generation Approach

Traditional cloud GAN models take 30–60s, breaking classroom interactivity. This platform implements a fast local synthesizer (src/avatar_service.py):

1. Uses OpenCV Haar Cascades to detect facial landmarks and the oral cavity on the educator portrait.
2. Extracts the audio volume envelope across 16 fps time slices to compute mouth opening ratios.
3. Applies affine mouth transformations with linear interpolation (cv2.INTER_LINEAR) and natural head nodding.
4. Muxes audio and encodes H.264 MP4 in **under 1.9 seconds**.
5. Served via HTTP 206 Partial Content range requests for instant browser streaming.

13. APIs and Third-Party Services

- **Groq Cloud API:** High-speed LLM inference (Llama 3.3 70B & 8B) and STT (Whisper Large v3).
- **Microsoft Edge-TTS:** Neural speech audio generation.
- **Pollinations AI:** Educator persona portrait synthesis.
- **LangSmith:** LLM call tracing, observability, and latency monitoring.
- **ChromaDB:** Local embedded vector database for document grounding.

14. Known Limitations

- **Neural Voice Connectivity:** Edge-TTS requires active outbound internet; falls back to standard gTTS if unavailable.
 - **2D Facial Animation:** Prioritizes sub-2-second generation speed over compute-heavy 3D mesh rendering.
 - **Database Scope:** Uses SQLite for local persistence; easily upgradeable to PostgreSQL for large-scale enterprise clustering.
 - **Browser Autoplay Policy:** Modern browsers prevent unmuted video autoplay until the user interacts with the page once.
-